Theorem[section] [theorem]Lemma [theorem]Corollary [theorem]Definition
Problem[section]

Overfitting in Energy-Based Models

Implicit (early stopping) spectral regularization in EBMs

Enrico Yao-Bate, Hanming Ye

December 9, 2025

Motivation

In machine learning, we want our model to generalize, not memorize. This is especially important in **generative modeling**: we want models to produce new samples that resemble the data, i.e., match its underlying probability distribution, rather than simply copying the training set.

Motivation: a (non-generative) setting we understand well

High-dimensional linear regression

- Supervised setting: we observe pairs (x_i, y_i) and want to predict $y \in \mathbb{R}$ from design matrix $X \in \mathbb{R}^{n \times p}$ via the model $y = X\beta + \varepsilon$
- Two basic quantities:
  - *training error*: how well the model fits the observed (x_i, y_i)
  - *test error*: expected error on *new* (x, y), our standard quantitative measure of how well the model **generalizes**.
- When the number of features p is comparable to the sample size n, a linear model can perfectly fit the training data (zero training error) but still have large test error (**overfitting**).

Motivation: a (non-generative) setting we understand well

- In this setting there is a detailed mathematical theory (Hastie–Montanari–Rosset–Tibshirani, 2019):
  - it describes how the gap between training and test error depends on the geometry of the features and the ratio p/n
  - it shows how to reduce test error by **regularizing** the model, either explicitly (ridge) or implicitly (early stopping for gradient descent)
  - underlying all of this is a *spectral* viewpoint: different directions in feature space (eigenvectors of $X^\top X$) have different noise levels and are regularized differently.

Background: high-dimensional regression

Set-up

- Linear model: $y = X\beta^{+\varepsilon}$, $X \in \mathbb{R}^{n\times p}$.
- Let $X^\top X = U\Sigma V^\top$ be the SVD with singular values σ_j.

1. Explicit regularization (ridge)

- Ridge estimator:

$$\hat{\beta}_\lambda = (X^\top X + n\lambda I_p)^{-1} X^\top y.$$

- In the eigenbasis of $X^\top X$,

$$\hat{\beta}_\lambda = \sum_{j=1}^{\min\{n,p\}} \frac{\sigma_j}{\sigma_j^2 + n\lambda} \left(u_j^\top y\right) v_j,$$

so directions with small σ_j^2 (noisy directions) are shrunk more.

Background: high-dimensional regression

2. **Implicit regularization (early stopping)**
   - Gradient descent on the least-squares loss $\frac{1}{2n}\|y - X\beta\|_2^2$:

$$\beta^{(k+1)} = \beta^{(k)} + \eta X^\top (y - X\beta^{(k)}), \quad \beta^{(0)} = 0.$$

   - One can show (for suitable step size η) that $\beta^{(k)}$ can also be written as a spectral filter:

$$\beta^{(k)} = \sum_j f_k(\sigma_j^2)\,(u_j^\top y)\,v_j, \quad f_k(\sigma_j)^2 = \frac{1 - (1 - \eta\sigma_j^2)^k}{\sigma_j}.$$

   For finite k, high-variance directions are updated more than noisy ones: **early stopping acts as implicit spectral regularization**.

Motivation: our question

Is there an analogous implicit regularization phenomena for energy-based models, and can we characterize it?

Motivation: our setting

Energy-Based Model (EBM):

- We are given samples $x_1, \ldots, x_n \in \mathbb{R}^d$ from an unknown density $\rho^{(x)}$.
- An EBM proposes an energy function $U_\theta : \mathbb{R}^d \to \mathbb{R}$ and then defines a Boltzmann–Gibbs density

$$\rho_\theta(x) = \frac{1}{Z_\theta} \exp(-U_\theta(x)), \qquad Z_\theta = \int_{\mathbb{R}^d} e^{-U_\theta(x)}\, dx.$$

Any smooth density can be written this way for a suitable U_θ.

Background: Energy-Based Models

EBMs vs other generative models (Carbone, 2025):

- They are especially attractive when we care about **interpretable energy landscapes** or couplings (e.g. Gaussian graphical models, inverse Ising, scientific applications).
- EBMs provide an explicit (up to Z_θ) probability density and connect naturally to statistical physics
- Energy functions are composable

Background: Energy-Based Models

Typical training methods (Song & Kingma, 2021):

- **Maximum likelihood / cross-entropy minimization**: minimize $H(\rho^*, \rho_\theta)$ or equivalently $\mathrm{KL}(\rho^* \| \rho_\theta)$, which leads to gradients

$$\partial_\theta H(\rho^*, \rho_\theta) = \mathbb{E}_{\rho^*}[\partial_\theta U_\theta] - \mathbb{E}_{\rho_\theta}[\partial_\theta U_\theta].$$

- Estimating $\mathbb{E}_{\rho_\theta}[\cdot]$ requires sampling from ρ_θ at each step (Langevin, Metropolis–Hastings, etc.), leading to well-known training difficulties.
- Sampling-free surrogates (**score matching**, **noise contrastive estimation**, etc.) avoid Z_θ but lose exact normalization.

Background: overfitting in EBMs

Setting (Catania–Decelle–Furtlehner–Seoane, 2025)

- Focus on simple but important EBMs:
  - **Gaussian EBM**:
    $$E(x|J) = \frac{1}{2}x^\top Jx$$
    with $J \succ 0$, so $p(x|J)$ is Gaussian with covariance $C = J^{-1}$.
  - **Pairwise Ising/Boltzmann machine**
- They assume data are generated from a ground-truth model (J^*) and study how gradient-based learning behaves when the number of samples M and dimension N grow.

Background: overfitting in EBMs

Model

- $p(x \mid J) \propto (\det J)^{1/2} \exp\big(-\frac{1}{2}x^\top J x\big)$, with $C = J^{-1}$ our covariance
- Empirical covariance from M samples:

$$\hat{C}_M = \frac{1}{M}\sum_{\mu=1}^{M} x^{(\mu)} x^{(\mu)\top}.$$

Maximum likelihood

- Log-likelihood (up to constants):

$$L(J) = -\tfrac{1}{2}\mathrm{Tr}(J\hat{C}_M) + \tfrac{1}{2}\log\det J.$$

- Gradient ascent with step size η:

$$\nabla_J L(J) = \tfrac{1}{2}(J^{-1} - \hat{C}_M), \quad J_{t+1} = J_t + \eta(J_t^{-1} - \hat{C}_M).$$

Background: overfitting in EBMs

Theorem 1 (Spectral gradient flow in Gaussian EBMs)

There exist scalar functions $\lambda_\alpha : [0, \infty) \to (0, \infty)$ *such that* $J_t = U \Lambda_t U^\top$, *where* $\Lambda_t = \mathrm{diag}(\lambda_1(t), \ldots, \lambda_N(t))$ *and for each eigenmode* α*:*

1. $\lambda_\alpha(t)$ *evolves independently according to the scalar ODE*

$$\dot{\lambda}_\alpha(t) = \frac{1}{\lambda_\alpha(t)} - \hat{c}_\alpha,$$

   with unique stable fixed point $\lambda_\alpha^\star = \frac{1}{\hat{c}_\alpha}$.

2. *Linearizing around* $\lambda_\alpha^\star$, *the deviation* $\delta_\alpha(t) := \lambda_\alpha(t) - \lambda_\alpha^\star$ *satisfies*

$$\dot{\delta}_\alpha(t) \approx -\hat{c}_\alpha^2 \, \delta_\alpha(t), \quad \Longrightarrow \quad \delta_\alpha(t) \approx \delta_\alpha(0) \, e^{-\hat{c}_\alpha^2 t}.$$

   In particular, the convergence timescale of mode α *is* $\tau_\alpha \sim \frac{1}{c_\alpha^2}$.

Background: overfitting in EBMs

Interpretation: Gaussian EBM training is *spectral*: each eigenmode of $\hat{C}_M$ is learned independently via a scalar ODE, and strong eigenmodes (large $\hat{c}_\alpha$) are fit much faster than weak/noisy ones. Stopping the flow at a finite time t therefore acts as an **implicit spectral regularizer** on J_t.

Motivation: regularization

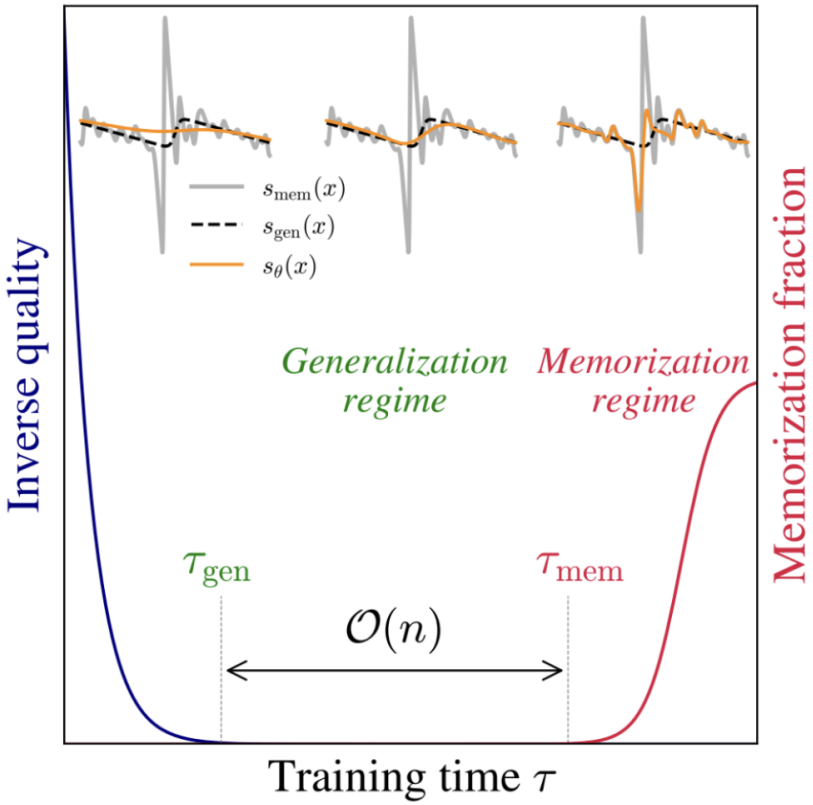
Architecture
(eg. inductive bias of MLPs)

Data
(eg. low-dimensional manifolds)

Training dynamics

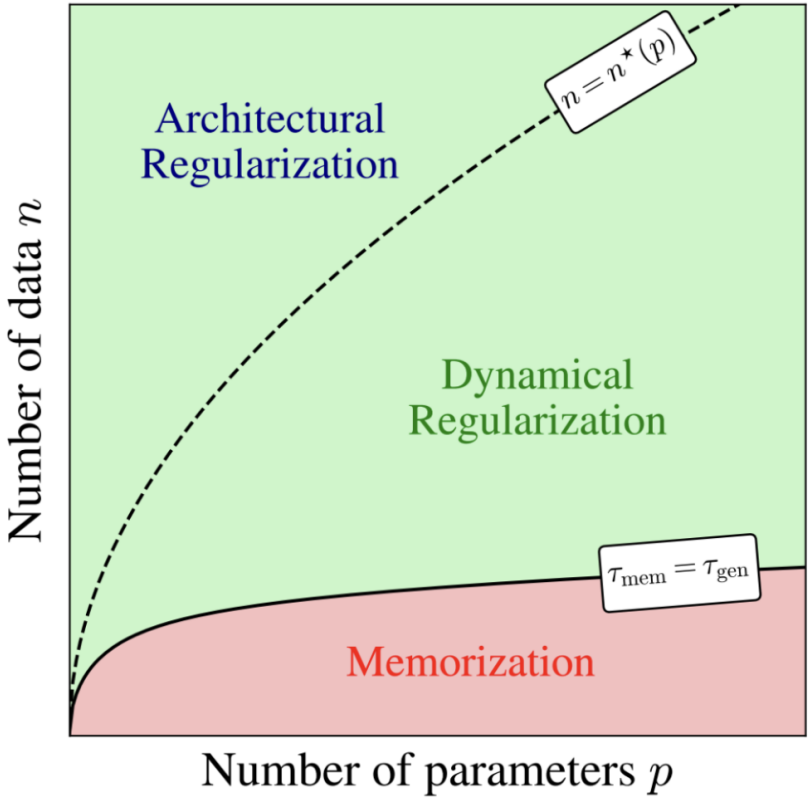
Past work has examined this for diffusion;
we aim to extend to EBMs

Implicit Regularization (in diffusion models)



Bonnaire et. al, 2025

Implicit Regularization (in diffusion models)



Bonnaire et. al, 2025

Theory roadmap

- Step 1: Define a random-feature score model and score-matching loss.

Theory roadmap

- Step 1: Define a random-feature score model and score-matching loss.
- Step 2: Derive gradient flow for the second-layer weights A.

Theory roadmap

- Step 1: Define a random-feature score model and score-matching loss.
- Step 2: Derive gradient flow for the second-layer weights A.
- Step 3: Solve this flow in closed form $\Rightarrow$ timescales = inverse eigenvalues of a matrix U.

Theory roadmap

- Step 1: Define a random-feature score model and score-matching loss.
- Step 2: Derive gradient flow for the second-layer weights A.
- Step 3: Solve this flow in closed form $\Rightarrow$ timescales = inverse eigenvalues of a matrix U.
- Step 4: Use random matrix theory to compute the spectrum of U

Random feature score model

- Fix a diffusion time t and consider the score at that time.

Random feature score model

- Fix a diffusion time t and consider the score at that time.
- Score model:

$$s_A(x) = \frac{A}{\sqrt{p}}\,\sigma\left(\frac{Wx}{\sqrt{d}}\right),$$

where

  - $W \in \mathbb{R}^{p\times d}$: i.i.d. Gaussian, frozen
  - $A \in \mathbb{R}^{d\times p}$: trainable
  - σ acts elementwise (e.g. tanh).

Score-matching loss (train and test)

- At fixed t, consider a rescaled score-matching loss:

$$L_{\text{train}}(A) = \frac{1}{d}\,\widehat{\mathcal{L}}_{\text{SM}}(A) \quad \text{on training samples.}$$

Score-matching loss (train and test)

- At fixed t, consider a rescaled score-matching loss:

$$L_{\text{train}}(A) = \frac{1}{d}\,\widehat{\mathcal{L}}_{\text{SM}}(A) \quad \text{on training samples.}$$

- Define the test loss $L_{\text{test}}(A)$ the same way but with an expectation over the ground-trth $x \sim P_X$.

Dynamics of training

- Study how A evolves under gradient descent on L_{train}.

Dynamics of training

- Study how A evolves under gradient descent on L_{train}.
- In the high-dimensional limit and small step size, gradient descent $\rightarrow$ gradient flow (continuous-time ODE).

Dynamics of training

- Study how A evolves under gradient descent on L_{train}.
- In the high-dimensional limit and small step size, gradient descent $\rightarrow$ gradient flow (continuous-time ODE).
- Realize that the gradient flow is linear in A and only depends on A through a matrix U.

Dynamics of training

- Study how A evolves under gradient descent on L_{train}.
- In the high-dimensional limit and small step size, gradient descent $\rightarrow$ gradient flow (continuous-time ODE).
- Realize that the gradient flow is linear in A and only depends on A through a matrix U.
- This lets us solve $A(\tau)$ explicitly and see how eigenvalues of U influence the training timescales.

Dynamics of training

- Discrete-time update:

$$A^{(k+1)} = A^{(k)} - \eta \nabla_A L_{\text{train}}(A^{(k)}).$$

Dynamics of training

- Discrete-time update:

$$A^{(k+1)} = A^{(k)} - \eta \nabla_A L_{\text{train}}(A^{(k)}).$$

- Rescale time:

$$\tau = \frac{k\eta}{d^2},$$

and let $\eta \to 0$.

Dynamics of training

- Discrete-time update:

$$A^{(k+1)} = A^{(k)} - \eta \nabla_A L_{\text{train}}(A^{(k)}).$$

- Rescale time:

$$\tau = \frac{k\eta}{d^2},$$

  and let $\eta \to 0$.
- In this limit, we obtain a gradient flow:

$$\dot{A}(\tau) = -d^2 \nabla_A L_{\text{train}}(A(\tau)).$$

Dynamics of training

- Discrete-time update:

$$A^{(k+1)} = A^{(k)} - \eta \nabla_A L_{\text{train}}(A^{(k)}).$$

- Rescale time:

$$\tau = \frac{k\eta}{d^2},$$

  and let $\eta \to 0$.
- In this limit, we obtain a gradient flow:

$$\dot{A}(\tau) = -d^2 \nabla_A L_{\text{train}}(A(\tau)).$$

- Next we compute $\nabla_A L_{\text{train}}$ explicitly.

Expressing the training loss

- Can expand the score-matching objective and write

$$L_{\text{train}}(A) = 1 + \frac{\Delta t}{d}\,\mathrm{Tr}\big(A^\top \sqrt{p}\,A\sqrt{p}\,U\big) + \frac{2\sqrt{\Delta t}}{d}\,\mathrm{Tr}\big(A\sqrt{p}\,V\big),$$

where $\Delta t = 1 - e^{-2t}$.

Expressing the training loss

- Can expand the score-matching objective and write

$$L_{\text{train}}(A) = 1 + \frac{\Delta t}{d}\,\mathrm{Tr}\big(A^\top \sqrt{p}\,A\sqrt{p}\,U\big) + \frac{2\sqrt{\Delta t}}{d}\,\mathrm{Tr}\big(A\sqrt{p}\,V\big),$$

where $\Delta t = 1 - e^{-2t}$.

- Here

$$U = \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}_\xi\Big[\sigma\big(\tfrac{Wx_\nu^t(\xi)}{\sqrt{d}}\big)\sigma\big(\tfrac{Wx_\nu^t(\xi)}{\sqrt{d}}\big)^\top\Big],$$

$$V = \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}_\xi\Big[\sigma\big(\tfrac{Wx_\nu^t(\xi)}{\sqrt{d}}\big)\,\xi^\top\Big],$$

and $x_\nu^t(\xi)$ is the noised data at time t.

Expressing the training loss

- Can expand the score-matching objective and write

$$L_{\text{train}}(A) = 1 + \frac{\Delta t}{d}\,\mathrm{Tr}\big(A^\top \sqrt{p}\,A\sqrt{p}\,U\big) + \frac{2\sqrt{\Delta t}}{d}\,\mathrm{Tr}\big(A\sqrt{p}\,V\big),$$

  where $\Delta t = 1 - e^{-2t}$.

- Here

$$U = \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}_\xi\Big[\sigma\big(\tfrac{Wx_\nu^t(\xi)}{\sqrt{d}}\big)\sigma\big(\tfrac{Wx_\nu^t(\xi)}{\sqrt{d}}\big)^\top\Big],$$

$$V = \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}_\xi\Big[\sigma\big(\tfrac{Wx_\nu^t(\xi)}{\sqrt{d}}\big)\,\xi^\top\Big],$$

  and $x_\nu^t(\xi)$ is the noised data at time t.

- U is the key object we will examine with (familiar techniques) like RMT.

Gradient of the loss

- Differentiate L_{train} w.r.t. A:

$$\nabla_A L_{\text{train}}(A) = \frac{2\Delta t}{d} ApU + \frac{2\sqrt{\Delta t}}{d}\sqrt{p}\, V^\top.$$

Gradient of the loss

- Differentiate L_{train} w.r.t. A:

$$\nabla_A L_{\text{train}}(A) = \frac{2\Delta t}{d} ApU + \frac{2\sqrt{\Delta t}}{d}\sqrt{p}\, V^\top.$$

- We can solve this ODE

Gradient of the loss

- Differentiate L_{train} w.r.t. A:

$$\nabla_A L_{\text{train}}(A) = \frac{2\Delta t}{d} ApU + \frac{2\sqrt{\Delta t}}{d}\sqrt{p}\, V^\top.$$

- We can solve this ODE
- Some algebra later...

Gradient of the loss

- Differentiate L_{train} w.r.t. A:

$$\nabla_A L_{\text{train}}(A) = \frac{2\Delta t}{d} ApU + \frac{2\sqrt{\Delta t}}{d}\sqrt{p}\, V^\top.$$

- We can solve this ODE
- Some algebra later...
-

$$\frac{A(\tau)}{\sqrt{p}} = -\frac{1}{\sqrt{\Delta t}} V^T U^{-1} + \left(\frac{1}{\sqrt{\Delta t}} V^T U^{-1} + \frac{A_0}{\sqrt{p}}\right) e^{-\frac{2\Delta t}{\psi_p} U\tau}.$$

Timescales from eigenvalues

- In eigenbasis of U, each mode follows

$$a_i(\tau) \sim e^{-\frac{2\Delta t}{\psi_p}\lambda_i \tau}$$

Timescales from eigenvalues

- In eigenbasis of U, each mode follows

$$a_i(\tau) \sim e^{-\frac{2\Delta t}{\psi_p}\lambda_i \tau}$$

- So the characteristic timescale of mode i is

$$\tau_i \sim \frac{\psi_p}{\Delta t\, \lambda_i}.$$

Timescales from eigenvalues

Evicence suggests that training dynamics for *energy-based models* follow a similar bias

- (Catania et al, 2025) Gaussian EBM: in the eigenbasis of the empirical covariance $C^{(M)}$, each coupling mode J_α evolves independently

$$\tau \frac{dJ_\alpha}{dt} = \frac{1}{J_\alpha} - \hat{c}_\alpha^{(M)}, \qquad J_\alpha(t) - J_\alpha(\infty) \sim e^{-(\hat{c}_\alpha^{(M)})^2 t/\tau},$$

  so the relaxation time scales as $t_{\mathrm{rel},\alpha} \propto (\hat{c}_\alpha^{(M)})^{-2}$.

The two bulks

- We know training timescales are controlled by the eigenvalues of U.

The two bulks

- We know training timescales are controlled by the eigenvalues of U.
- Now we need the eigenvalue distribution of U.

The two bulks

- We know training timescales are controlled by the eigenvalues of U.
- Now we need the eigenvalue distribution of U.
- Strategy:
  - Use Gaussian Equivalence.
  - Use the (familiar) resolvent.

Gaussian Equivalence Principle (GEP)

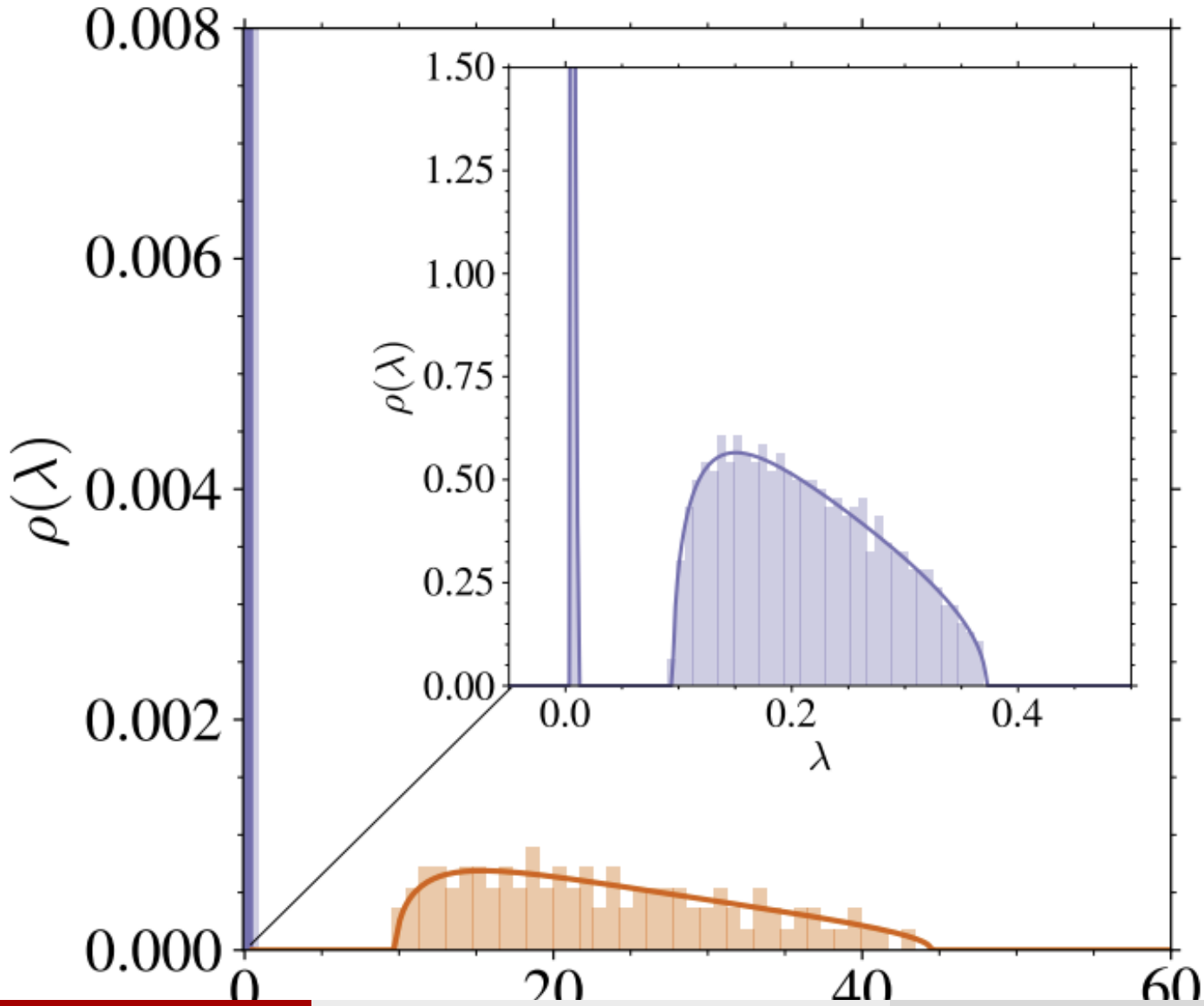
- Replace the nonlinear activation function by a linear term and a nonlinear term that acting as noise

Gaussian Equivalence Principle (GEP)

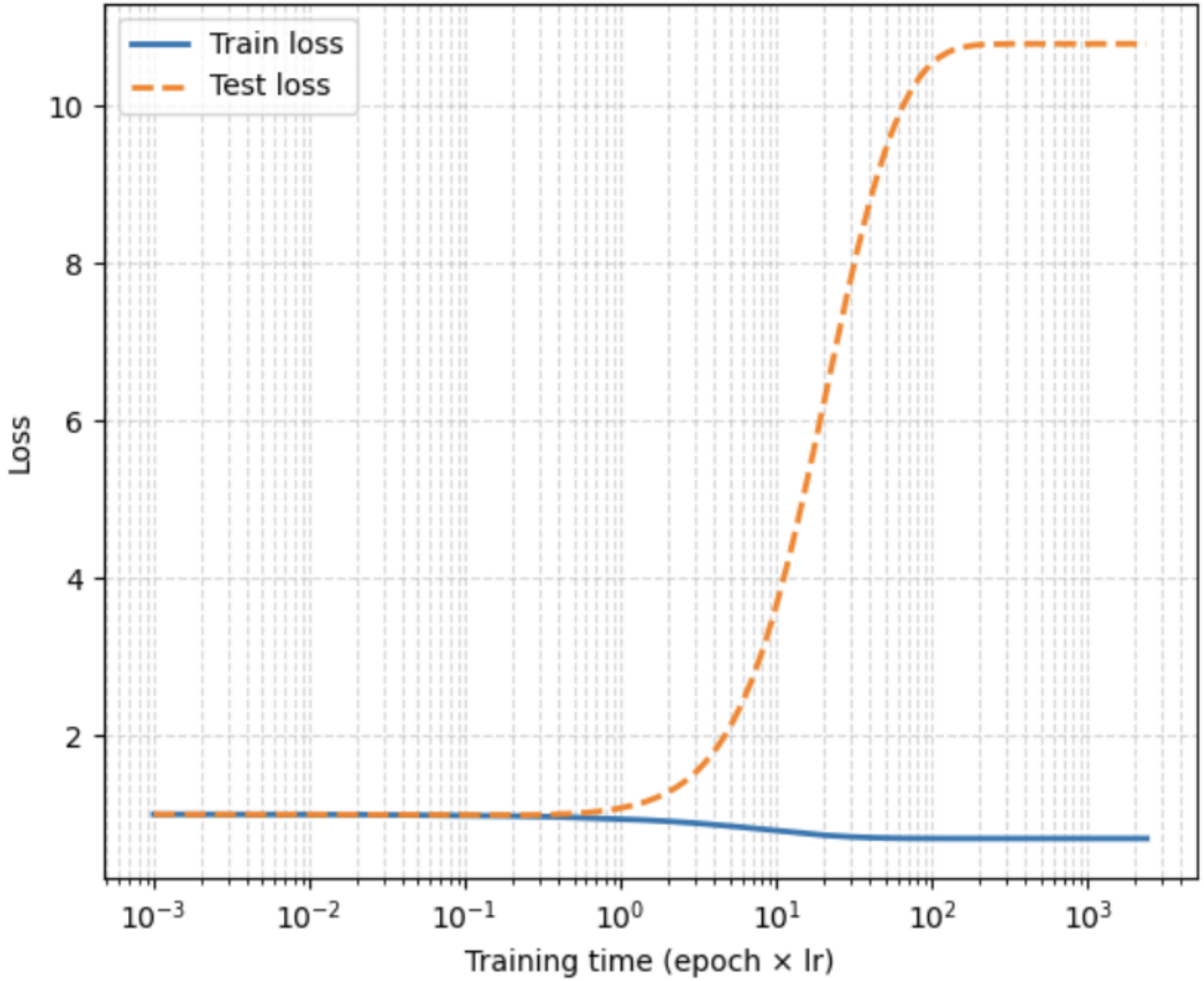
- Replace the nonlinear activation function by a linear term and a nonlinear term that acting as noise
- U has same spectrum as:

$$U_{\text{GEP}} = \frac{1}{n} GG^\top + b_t^2 \frac{1}{d} WW^\top + s_t^2 I_p,$$

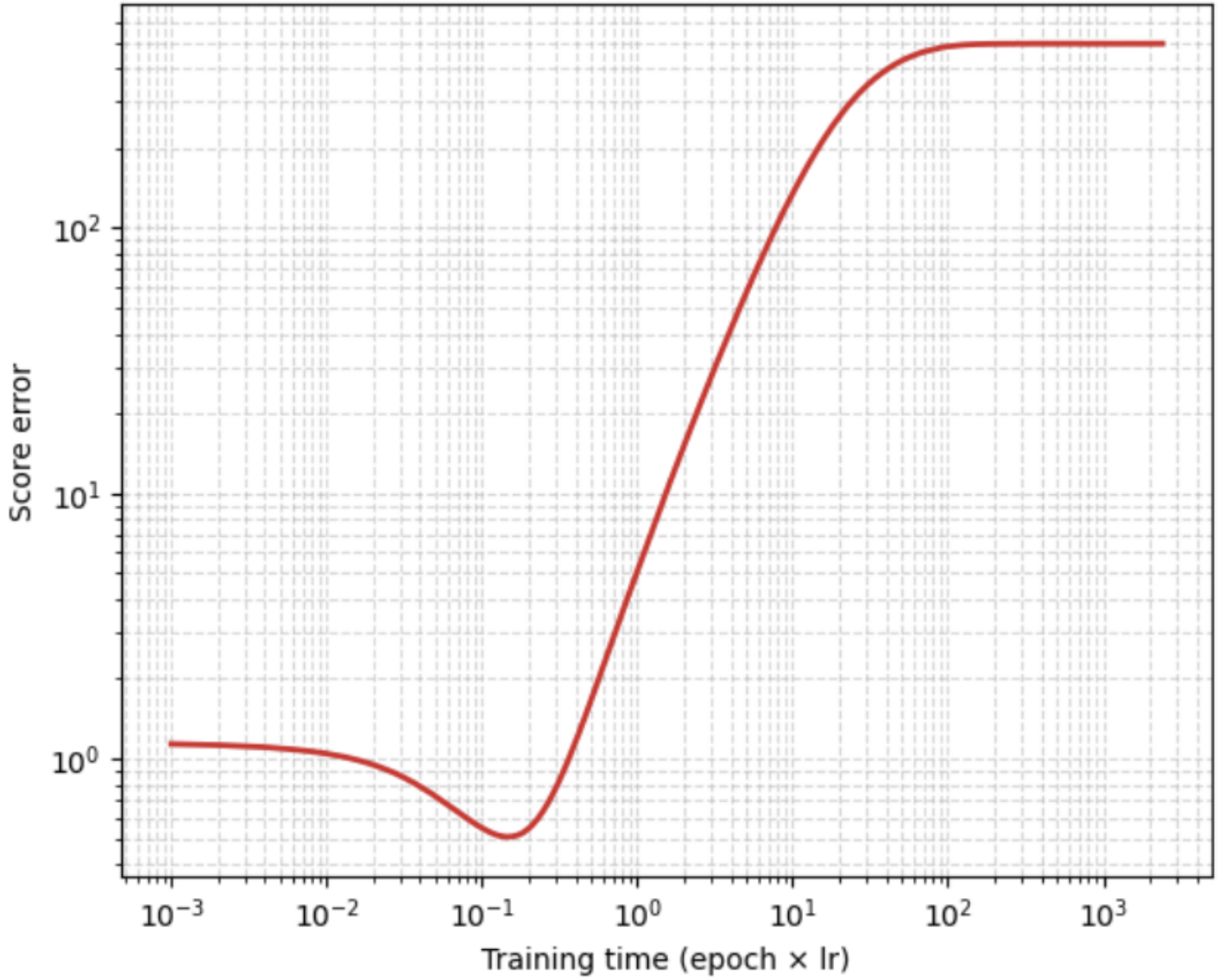
Spectral density



For EBMs...


Train loss
Test loss
Loss
10
8
6
4
2
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Training time (epoch × lr)

For EBMs...


Score error
10^2
10^1
10^0
10^{-3}
10^{-2}
10^{-1}
10^0
10^1
10^2
10^3
Training time (epoch × lr)

References

- G. Catania, A. Decelle, C. Furtlehner, B. Seoane. *A Theoretical Framework for Overfitting in Energy-Based Modeling.* ICML 2025. arXiv:2501.19158.
- T. Bonnaire, R. Urfin, G. Biroli, M. Mézard. *Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training.* NeurIPS 2025 (to appear). arXiv:2505.17638.
- D. Carbone. *Hitchhiker's Guide on the Relation of Energy-Based Models with Other Generative Models, Sampling and Statistical Physics: A Comprehensive Review.* Trans. Mach. Learn. Res., 2025. arXiv:2406.13661.
- Y. Song, D. P. Kingma. *How to Train Your Energy-Based Models.* Tutorial chapter in *Probabilistic Machine Learning: Advanced Topics*, 2021. arXiv:2101.03288.
- T. Hastie, A. Montanari, S. Rosset, R. J. Tibshirani. *Surprises in High-Dimensional Ridgeless Least Squares Interpolation.* Ann. Statist. 50(2):949–986, 2022. arXiv:1903.08560.